Algebraic Topology

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Chapter 1

Fundamental Group

1. Homotopy

Definition 1.1 (Homotopic maps)**.** Let $f, f' : X \to Y$ be two maps. We say that f is homotopic to f' if there exists a continuous map $F : X \times I \to Y$ such that:

$$\forall x \in X : F(x, 0) = f(x) \text{ and } F(x, 1) = f'(x)$$

and the map f is said to be homotopic to f, written $f \simeq f'$.

Definition 1.2 (Path Homotopy)**.** Two paths $f, g : [0, 1] \to X$ with same start and end points,say a, b respectively are said to be path homotopic if there exists a homotopy $F : I \times I \to X$ such that

$$\forall t \in [0, 1] : F(0, t) = a, F(1, t) = b$$

and we write $\simeq_p$

Ex 1.1. The relations $\simeq$ and $\simeq_p$ are equivalence relations

Definition 1.3 (Product/Concatenation of paths)**.** Let $f, g : [0, 1] \to X$ be paths in X, such that $f(1) = g(0)$. We define the product $f \star g$ of the paths to by

$$f \star g = \begin{cases} f(2t) & t \leq \frac{1}{2} \\ g(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

Ex 1.2. The product operation defines a well defined operation on path homotopy classes. I.e, if $[f]$ denotes the path homotopy class of f, then $[g] \star [h] := [g \star h]$ is a well defined operation on path homotopy classes

Ex 1.3. Furthermore, find an "identity" path, and show that the product operation satisfies the group axioms

2. The Fundamental Group

A path starting and ending at a point is said to be a loop based at that point.

Definition 2.1. Let X be a topological space, and let $x_0 \in X$. The set of all path homotopy classes of loops based at x_0 is a group, denoted $\pi_1(X, x_0)$ and called the first fundamental group (or just the fundamental group) at x_0.

Theorem 2.2. *Let α be a path from x_0 to x_1 and define a map*

$$\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$$

by

$$[f] \mapsto [\alpha]^{-1} \star [f] \star [\alpha]$$

where $[\alpha]^{-1}$ is the path homotopy class of the reverse path from x_1 to x_0. The map $\hat{\alpha}$ is a group homomorphism. Furthermore, it is an isomorphism.

Proof. exercise. □

Note that the maps $\hat{\alpha}$ are not canonical isomorphism, as they reqire the choice of a path α. Also, when $x_0 = x_1$, $\hat{\alpha}$ is an inner automorphism.

Corollary 2.3. *If X is path connected and $x_0, x_1 \in X$, we have that $\pi_1(X, x_0) \cong \pi_1(X, x_1)$. So it makes sense to talk about the fundamental group of a path connected space.*

Definition 2.4 (Simply connected)**.** A path connected space X is said to be simply connected if for some $x_0 \in X$, $\pi_1(X, x_0) = 0$ and hence for every $x \in X$.

Definition 2.5 (Pointed Top)**.** The category consisting of pairs (X, x_0) as objects and continuous maps $f : X \to Y$ such that $f(x_0) = y_0$ as morphisms between (X, x_0) to (Y, y_0) is called the category of pointed topological sapces, denoted $\mathbf{Top}_\star$.

Definition 2.6 (Induced homomorphism)**.** Let $\phi : (X, x_0) \to (Y, y_0)$ be a map of pointed topological spaces. Define the map

$$\phi_\star : \pi_1(X, x_0) \to \pi_1(Y, y_0)$$

by $[f] \mapsto [\phi \circ f]$. This map is said to be the homomorphism induced by ϕ.

Ex 2.1. Show that the induced homomorphism is indeed a group homomorphism

I Functorial properties of the induced homomorphism

Theorem 2.7. *If $f : (X, x_0) \to (Y, y_0)$ and $g : (Y, y_0) \to (Z, z_0)$ are maps of pointed topological spaces, then $(g \circ f)_\star = g_\star \circ f_\star$. Furthermore, if $i : (X, x_0) \to (X, x_0)$ is the identity map, then $i_\star = \mathrm{id}_{\pi_1(X, x_0)}$.*

Proof. exercise □

All of this is to say that taking the fundamental group at a point is a functor $\mathbf{Top}_\star \to \mathbf{Grp}$.

Corollary 2.8. *If $h : (X, x_0) \to (Y, y_0)$ is an isomorphism, then $h_\star : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ is an isomorphism.*

(The above holds for any functor, but we state it explicitly anyways.)

3. Covering Spaces

Definition 3.1. Let $p : E \to B$ be a surjective map of topological spaces. The open set U of B is said to be **evenly covered** by B if $p^{-1}(U)$ can be written as the disjoint union of open sets V_α such that the restriction of p to V_α is a homeomorphism onto U. The collection $\{V_\alpha\}$ will be called a partition of $p^{-1}(U)$ into **slices**

Definition 3.2 (Covering space)**.** Let $p : E \to B$ be a surjective map of topological sapces. If for every $b \in B$, there is an open $U \supset \{b\}$ such that U is evenly covered by p, then E is said to be a covering space for B and p is said to be a **covering map** .

That is to say, if B is locally evenly covered by p, then E is a covering space for B.

Ex 3.1. Show that the above definition of covering space is equivalent to the definition of covering space derived from the following definition for evenly covered set:

Definition 3.3 (Covering Space, alternate definition)**.** $p : E \to B$ surjective map of topological spaces, $U \subseteq B$ is said to be evenly covered if there exists a set A with the discrete topology such that there exists a homeomorphism $\varphi : U \times A \to p^{-1}(U)$ making the following diagram commutative:

$$\begin{tikzcd} U \times A \arrow[rr, "\varphi", "\simeq"'] \arrow[dr, "\pi_1"'] & & p^{-1}(U) \arrow[dl, "p"] \\ & U & \end{tikzcd}$$

Example 1. The real line with the map $x \mapsto (\cos(x), \sin(x))$ is a covering map $\mathbb{R} \to S^1$, and $\mathbb{R}$ is thus a covering space for S^1.

A space is always a covering space of itself, with covering map identity. However, in the case of Example 1, there are more maps than just the identity. Namely those given by $(\cos(x), \sin(x)) \mapsto (\cos(nx), \sin(nx))$, one for each integer n. Now, visually speaking, the fundamental group of S^1 seems to be what? Does this hint to some sort of Galois Correspondence for convering maps and subgroups of the fundamental group given sufficiently "good" conditions?

Ex 3.2. Show that covering maps are open maps.

Ex 3.3. Let $p : E \to B$ be a covering map, and let $B_0 \subseteq B$ be a subspace. Show that $p|_{p^{-1}(B_0)} : p^{-1}(B_0) \to B_0$ is a covering map.

Ex 3.4. product of covering maps is a covering map

Ex 3.5. Define a figure 8 to be two non degenerate circles meeting at a single common point. Show that the 'infinite grid' $\mathbb{R} \times \mathbb{Z} \cup \mathbb{Z} \times \mathbb{R}$ is a covering space of the figure 8. (Hint: You should be able to show this without needing to explicitly write down the map p by using the previous exercise(s). You can also write down the map explicitly if you want, simply use the maps in Example 1)

How do coverings relate to the fundamental group? we make this connection using lifting.

Definition 3.4. Let $p : E \to B$ be a covering map. A lift of $f : X \to B$ along p is a map $\tilde{f}$ such that $p \circ \tilde{f} = f$.

I Lifting Properties of Covering Spaces

Theorem 3.5. *Let $p : E \to B$ be a covering map, and let $f : [0, 1] \to B$ be a path with $f(0) = b$. If $e \in p^{-1}(b)$, then there is a unique lift $\tilde{f} : [0, 1] \to E$ with $\tilde{f}(0) = e$.*

Proof. Since p is a covering map, there exists an open cover $\{U_\alpha\}_{\alpha \in J}$ of B such that each U_α is evenly covered by p. Now, note that $\{f^{-1}(U_\alpha)\}_{\alpha \in J}$ is an open cover of $[0, 1]$, and by the lebesgue number lemma, there is a $\delta > 0$ with the property that for any $x \in [0, 1]$, we have $(x, x + \delta) \subseteq f^{-1}(U_\alpha)$ for some $\alpha \in J$. Hence we can find a finite subdivision $0 = s_0 < s_1 < \cdots < s_n = 1$ such that $f([s_i, s_{i+1}]) \subseteq U_\alpha$ for some $\alpha \in J$.

Define $\tilde{f}(0) = e$, and assume that $\tilde{f}$ has been continuously defined on $[0, s_i]$ for some $i \geq 0$. We extend $\tilde{f}$ continuously to $[0, s_{i+1}]$. In order to do so, note that $f([s_i, s_{i+1}]) \subseteq U_\alpha$ for some $\alpha \in J$, and $p^{-1}(U_\alpha) = \bigsqcup_{j \in I} V_j \cdots (1)$ for some open V_j with $p|_{V_j} : V_j \to U_\alpha$ a homeomorphism for each $j \in I$. Pick the unique j with $\tilde{f}(s_i) \in V_j$. Then using (1) we have that $\tilde{f}(s) = p|_{V_j}^{-1}(f(s))$ is a continuous extension to $[0, s_{i+1}]$ by the pasting lemma. $\square$

The above theorem shows that paths lift uniquely to other paths in the covering space given the starting point of the lifted path.

Ex 3.6. Let $p : E \to B$ be a covering map and let $F : [0,1] \times [0,1] \to B$ be a continuous map with $F((0,0)) = b$, and let $e \in F^{-1}(b)$. Show that there exists a unique "lift" $\tilde{F} : [0,1] \times [0,1] \to E$ with $\tilde{F}((0,0)) = e$. (Hint: just repeat the previous proof, making necessary changes along the way.)

Ex 3.7. Use the previous exercise to show that homotopies (resp. path homotopies) "lift" to homotopies (resp. path homotopies) uniquely

Remark 1. Note that loops need not lift to loops (this should be clear by visualising the map in Example 1)

But, all hope is not lost. Given a loop and starting point for the lift, the ending point is uniquely determined. So, given $b_0 \in B$ and a point $e_0 \in p^{-1}(b_0)$ we have a map

$$\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$$

given by $\phi([f]) \mapsto \tilde{f}(1)$ where $\tilde{f}$ is the unique lift of f along p with $\tilde{f}(0) = e_0$. Note that this map is not canonical, since it depends on the choice of e_0. Furthermore, we need to show that this map is well defined.

Ex 3.8. Show that the lifting correspondence is well defined.

Definition 3.6. the map $\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$ is called the lifting correspondence. (I like to say lifting correspondence w.r.t e_0)

Theorem 3.7. *Let $p : E \to B$ be a covering map, and let $b_0 \in B$. Let $e_0 \in p^{-1}(b_0)$. If E is path connected, then the lifting correspondence*

$$\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$$

w.r.t e_0 is surjective. If E is simply connected, it is bijective.

Proof. exercise. □

Theorem 3.8. *We have the isomorphism*

$$\pi_1(S^1) \cong \mathbb{Z}$$

Proof. The map $t \mapsto (\cos(2\pi t), \sin(2\pi t))$ is a covering map, and we take the lifting correspondence $\phi : \pi_1(S^1) \cong \pi_1(S^1, (0,1)) \to p^{-1}((0,1)) = \mathbb{Z}$ w.r.t 0. This map is certainly bijective, since $\mathbb{R}$ is simply connected. We need to show that this map is a homomorphism. To that end, let $\phi([f]) = m, \phi([g]) = n$. Note that if $\tilde{f}, \tilde{g}$ are the unique lifts of f, g along p, then $\tilde{f} \star (m + \tilde{g})$ goes from 0 to $m + n$, and a simple computation shows that it is a lifting of $f \star g$ along p. Thus $\phi([f] \star [g]) = m + n = \phi([f]) + \phi([g])$. □

4. Consequences of $\pi_1(S^1) \cong \mathbb{Z}$

Definition 4.1. Let X be a topological space. For $A \subseteq X$, a retraction $r : X \to A$ is a map of topological spaces such that $r \circ i_A = id_A$ where $i_A : A \hookrightarrow X$ is the inclusion. If for a subspace A there exists such an r, we say that A is a retract of X.

Ex 4.1. If A is a retract of X, and $a_0 \in A$ then the induced map $(i_A)_\star : \pi_1(A, a_0) \to \pi_1(X, a_0)$ is an injection. (What does it say about $r_\star$?)

Lemma 4.2. *S^1 is not a retract of B^2. Here $B^2 = \{(x, y) : x^2 + y^2 \leq 1\}$*

Proof. Indeed, this is because there is no injection $\pi_1(S^1) = \mathbb{Z} \to \pi_1(B^2) = \{0\}$ □

Theorem 4.3 (Brouwer's fixed point theorem for 2 dimensions)**.** *Let $f : B^2 \to B^2$ be a continuous mapping. Then f has a fixed point.*

Proof. We assume that there is a map $f : B^2 \to B^2$ with no fixed points for contradcition. Thus, $\forall x \in B^2 : x \neq f(x)$, so we can pick $t_x > 0$ such that $|x + (f(x) - x)t_x| = 1$. Now define

$$h(x) = f(x) + t_x(x - f(x))$$

and note that t_x varies continuously with x, because it can be solved for using the quadratic formula. Thus h is continuous, and we also have that $h(x) = x$ for all $x \in S^1$. Thus we have a retraction $h : B^2 \to S^1$, a contradcition. □

Ex 4.2. prove the one dimensional version of the above theorem.

I Quotient Spaces

Definition 4.4. Let $f : X \to Y$ be a surjective function, where X is a topological space. The quotient topology on Y induced by X, f is the collection of subsets $V \subseteq Y$ such that $f^{-1}(V)$ is open in X.

Ex 4.3. verify that this is indeed a topology.

Definition 4.5. a surjective function $f : X \to Y$ is said to be a quotient map if Y has the quotient topology from X, f.

Remark 2. note that a quotient map is continuous.

Lemma 4.6. *Let $f : X \to Y$ be a function. Then the quotient topology on Y induced by X, f is the finest topology with respect to which f is continuous.*

Proof. exercise. □

Remark 3. A surjective function $f : X \to Y$ is a quotient map iff the following condition holds:
A subset K is closed in Y iff $f^{-1}(K)$ is closed in X

Lemma 4.7. *if $f : X \to Y$ is a surjective open or closed map, then f is a quotient map.*

Theorem 4.8. *Let $f : X \to Y$ be a quotient map. Let $B \subseteq Y$ and $A = f^{-1}(B) \subseteq X$ be subspaces and let $g : A \to B$ be the restricted map.*

1. *If A is open or closed, then g is a quotient map*

2. *If f is an open or closed map, then g is a quotient map.*

Proof. exercise □

Perhaps the most important property of quotient maps is the following:

Theorem 4.9. *Let $f : X \to Y$ be a quotient map, and let Z be a top. sp. Let $h : X \to Z$ be a function such that $h(x) = h(x')$ whenever $f(x) = f(x')$. Then h induces a unique function $g : Y \to Z$ such that $h = g \circ f$. The function g is continuous iff h is continuous. The function g is a quotient map iff h is a quotient map.*

These are not hard to prove, but they are results to keep in mind.

Corollary 4.10. *Let $h : X \to Z$ be a continuous surjective map. Let $\sim$ be the equivalence relation $x \sim y$ iff $h(x) = h(y)$, and let $X/\sim$. Give $X/\sim$ the quotient topology from the canonical surjection $\pi : X \to X/\sim$.*

1. *the map h induces a continuous bijective map $g : X/\sim \to Z$*

2. *the above map is a homeomorphism iff h is a quotient map.*

3. *If Z is hausdorff, so is $X/\sim$.*

5. Van - Kampen Theorem

Let X be a topological space, and U, V open subsets of X with the property that $X = U \cup V$, and $U \cap V \neq \emptyset$. The four inclusion maps:

$$\begin{tikzcd} & U \arrow[dr, hook, "j_1"] & \\ U \cap V \arrow[ur, hook, "i_1"] \arrow[dr, hook, "i_1"'] & & X \\ & V \arrow[ur, hook, "j_2"'] & \end{tikzcd}$$

induce fundamental group homomorphisms] By the universal property of the free product, there exists a homomorphism $\Phi : \pi_1(U,p) \star \pi_1(X,p)$ such that the following diagram commutes

$$\begin{tikzcd} & \pi_1(U,p) \arrow[dr, "(j_1)_\star"] & \\ \pi_1(U \cap V, p) \arrow[ur, "(i_1)_*"] \arrow[dr, "(i_1)_\star"'] & & \pi_1(X,p) \\ & \pi_1(V,p) \arrow[ur, "(j_2)_\star"'] & \end{tikzcd}$$

Theorem 5.1. *Let X be a top. sp. Suppose that $U, V \subseteq X$ are open with $X = U \cup V$, U and V path connected. Let $p \in U \cap V$, and define $C \subseteq \pi_1(U,p) \star \pi_1(V,p)$ by*

$$C = \{(i_\star \gamma)(j_\star \gamma) : \gamma \in \pi_1(U \cap V, p)\}$$

Then the map Φ is surjective, its kernel is the normal closure of C in $\pi_1(U,p) \star \pi_1(V,p)$. Therefore, $\pi_1(X,p) \cong (\pi_1(U,p) \star \pi_1(V,p))/\overline{C}$

Definition 5.2. Let f, h be group homomorphisms $H \to G_1$, $H \to G_2$ respectively. The amalgamated free product of G_1, G_2 along H is denoted by $G_1 \star_H G_2$, is the quotient group $(G_1 \star G_2)/\overline{C}$ where C is the set $\{f(g)h^{-1}(g) : g \in H\}$, thought of as a subset of $G_1 \star G_2$ by means of the usual inclusions of G_1, G_2 into the free product.

therefore, the van kampen theorem states that $\pi_1(X,p) \cong \pi_1(U,p) \star_{\pi_1(U \cap V, p)} \pi_1(V,p)$. I wont be providing proofs, since they will just be copy pastes of [Lee10] .

Example 2. The wedge sum $X_1 \vee \cdots \vee X_n$ of topological spaces is defined as the quotient of $\bigsqcup_j X_j$ by the equivalence relation generated by $p_1 \cdot \tilde{\ } \cdot \tilde{p}_n \cdot$. Let $\star$ be the point in $X_1 \vee \cdots \vee X_n$ that is the equivalence class of the points p_i.

Definition 5.3. a point $p \in X$ is said to be a nondegenerate base point if there exists a neighbourhood W of p that admits a strong deformation retract onto p.

Note for example, that every point in a manifold is a nondegenerate base point.

Lemma 5.4. *Suppose that $p_i \in X_i$ are nondegenerate base points for every i. Then $\star$ is a nondegenerate base point in $X_1 \vee \cdots \vee X_n$.*

Proof. Indeed, if W_i are neighbourhoods admitting strong deformation retracts H_i, then defining $H : (\bigsqcup_i W_i) \times I \to \bigsqcup_i W_i$ induces the desired strong deformation retract. $\square$

Theorem 5.5. *If $X_1, \cdots, X_n$ are spaces with nondegenerate base points p_i. The map*

$$\Phi : \bigstar_i \pi_1(X_i, p_i) \to \pi_1(X_1 \vee \cdots X_n)$$

induced by the inclusion $i_j : X_j \hookrightarrow X_1 \vee \cdots \vee X_n$ is an isomorphism.

Proof. We prove this for the case of two spaces. We want to be able to take $U = X_1, V = X_2$, but the problem is that these are not open. So, if W_i are neighbourhood of p_i admitting strong deformation retracts onto p_i, we consider $U_1 = q(X_1 \sqcup W_2), U_2 = q(X_2 \sqcup W_1)$ where q is the canonical map $X_1 \sqcup X_2 \to X_1 \vee X_2$. Further, the inclusions

$$\{\star\} \hookrightarrow U_1 \cap U_2 X_1 \hookrightarrow U_1, X_2 \hookrightarrow U_2$$

are homotopy equivalences. Furthermore, $U_1 \cap U_2$ is contractible hence simply connected, so that

$$\pi_1(U_1, \star) \star \pi_1(U_2, \star) \to \pi_1(X_1 \vee X_2, \star)$$

is an isomorphism by the Van Kampen Theorem. But since i_1, i_2 were homotopy equivalences, we have isomorphisms $\pi_1(X_i, p_i) \to \pi_1(U_i, \star)$. By induction, we are done. $\square$

Chapter 2

Singular Homology

1. The Singular Homology Group

Definition 1.1. A p simplex $\Delta_p \subseteq \mathbb{R}^p$ is defined to be equal to the convex set spanned by $\{e_i | i = 1, 2, .., p\} \cup \{0\}$.

Definition 1.2. Let X be a topological space. A singular p simplex on X is an continuous map $X \to \Delta_p$. The free abelian group generated by all singular p simplices on X is called the $p-$th singular chain group on X, written $C_p(X)$, and the elements of this group are called chains.

Definition 1.3. We define the $i-$th face of the singular p-simplex $F_{i,p} : \Delta_{p-1} \to \Delta_p$ as follows:

$$\begin{aligned} e_0 &\mapsto e_0 \\ e_1 &\mapsto e_1 \\ e_2 &\mapsto e_2 \\ &\vdots \\ e_{i-1} &\mapsto e_{i-1} \\ e_i &\mapsto e_{i+1} \\ e_{i+1} &\mapsto e_{i+2} \\ &\vdots \\ e_{p-1} &\mapsto e_p \end{aligned}$$

Definition 1.4. Let $\sigma : \Delta_p \to X$ be a singular p-simplex. Define a singular $p-1$-chain called the boundary of σ by:

$$\partial\sigma = \sum_{i=0}^{p} (-1)^i \sigma \circ F_{i,p}$$

Note that the map $\partial : C_p(X) \to C_{p-1}(X)$ is a group homomorphism, it is called the singular boundary operator, or just the boundary operator. Sometimes, we will also denote it by ∂_p.

Definition 1.5. A singular $p-$chain c is called a $p-$cycle if we have $\partial c = 0$. A singular $p-$chain c is called a $p-$boundary if there exists a $p+1-$chain b with $\partial b = c$.

Note that the set of all $p-$boundaries and cycles form groups, called $B_p(X), Z_p(X)$ respectively. Furthermore, we have that $\ker \partial_p = Z_p(X)$, and $\mathrm{Im}(\partial_{p+1}) = B_p(X)$.

Lemma 1.6. *If c is a singular chain, then $\partial(\partial c = 0)$.*

Proof. Note that it suffices to show the statement for singular $p-$simplices. We check the statement for $p \geq 2$. First we prove the following identity, for $i \leq j$:

$$F_{i,p} \circ F_{j,p} = F_{j,p} \circ F_{i-1,p}$$

Indeed, we have:

$$\begin{array}{cccccc}
F_{j,p-1} & F_{i,p} & & F_{i-1,p-1} & F_{j,p} \\
e_0 \mapsto & e_0 \mapsto e_0 & & e_0 \mapsto & e_0 \mapsto e_0 \\
\vdots & \vdots & & \vdots & \vdots \\
e_{j-1} \mapsto & e_{j-1} \mapsto e_{j-1} & & e_{j-1} \mapsto & e_{j-1} \mapsto e_{j-1} \\
e_j \mapsto & e_{j+1} \mapsto e_{j+1} & = & e_j \mapsto & e_j \mapsto e_{j+1} \\
\vdots & \vdots & & \vdots & \vdots \\
e_{i-2} \mapsto & e_{i-1} \mapsto e_{i-1} & & e_{i-2} \mapsto & e_{i-2} \mapsto e_{i-1} \\
e_{i-1} \mapsto & e_i \mapsto e_{i+1} & & e_{i-1} \mapsto & e_i \mapsto e_{i+1} \\
\vdots & \vdots & & \vdots & \vdots \\
e_{p-2} \mapsto & e_{p-1} \mapsto e_p & & e_{p-2} \mapsto & e_{p-1} \mapsto e_p
\end{array}$$

therefore:

$$\begin{aligned}
\partial(\partial\sigma) &= \sum_{i=0}^{p} (-1)^{i+j} \sigma \circ F_{i,p} \circ F_{j,p-1} \\
&= \sum_{0 \leq j <} \sum_{i \leq p} (-1)^{i+j} \sigma \circ F_{i,p} \circ F_{j,p-1} + \sum_{0 \leq i < j \leq p-1} (-1)^{i+j} \sigma \circ F_{i,p} \circ F_{j,p-1}
\end{aligned}$$

making the change of variables $j' = i, j = i' - 1$, we get:

$$\begin{aligned}
&= \sum_{0 \leq j <} \sum_{i \leq p} (-1)^{i+j} \sigma \circ F_{i,p} \circ F_{j,p-1} + \sum_{0 \leq j' <} \sum_{i' \leq p} (-1)^{i'+j'+1} \sigma \circ F_{j',p} \circ F_{i'-1,p-1} \\
&= 0 \quad \text{(using the identity we just proved)}
\end{aligned}$$

□

Therefore, we have that $B_p(X) \leq Z_p(X) \leq C_p(X)$. By convention, we take $C_{-1}(X) = 0$ for any X.

Definition 1.7. Define the $p-$th singular homology group of X by

$$H_p(X) = \frac{\ker(\partial_p)}{\operatorname{Im}(\partial_{p+1})}$$

Let $f : X \to Y$ be a map. Define $f_\sharp : C_p(X) \to C_p(Y)$ a group homomorphism by $f_\sharp(\sigma) = f \circ \sigma$ on the $p-$simplices.

Note that

$$f_\sharp(\partial\sigma) = f_\sharp\left(\sum_{i=0}^{p} (-1)^i \sigma \circ F_{i,p}\right) = \sum_{i=0}^{p} (-1)^i f \circ \sigma \circ F_{i,p} = \partial(f_\sharp\sigma)$$

therefore, $f_\sharp$ maps $Z_p(X)$ to $Z_p(Y)$, and $B_p(X)$ to $B_p(Y)$, so that it induces a map

$$f_* : H_p(X) \to H_p(Y).$$

Proposition 1.8. *Taking the $p-$th homology group is a functor* **Top** $\to$ **Ab. Grp**

Proof. Exercise. We have already proved the hard bits. □

Ex 1.1. What does abstract nonsense tell you about the map induced by f when it is a homeomorphism, and when f is a retraction?

2. A Brief Detour Into Homological Alegbra

Definition 2.1. Let $\{C_p\}_{p=0}^{\infty}$ be a sequence of abelian groups. The sequence

$$\cdots \to C_{p+1} \xrightarrow{\partial_{p+1}} C_p \xrightarrow{\partial_p} \cdots$$

is called a chain complex if $\forall p: \quad \partial_{p+1} \circ \partial_p = 0$

We denote a chain complex as above by C_*

The $p-$th homology group of C_* denoted $H_p(C_*)$ is defined similar to before.
Note that C_* is exact iff $H_p(C_*) = 0 \quad \forall p$.

Definition 2.2. Let C_*, D_* be chain complexes. $F : C_* \to D_*$ is called a chain map if for every $p > 0$, the following diagram commutes:

$$\begin{tikzcd}
\cdots \arrow[r] & C_{p+1} \arrow[r, "\partial"] \arrow[d, "F"] & C_p \arrow[r] \arrow[d, "F"'] & \cdots \\
\cdots \arrow[r] & D_{p+1} \arrow[r, "\partial"] & D_p \arrow[r] & \cdots
\end{tikzcd}$$

Definition 2.3. A sequence $\cdots \to C_* \to D_* \to E_* \cdots$ of chain complexes is said to be exact if for every $p > 0$, the sequence $\cdots C_p \to D_p \to E_p \to \cdots$ is exact.

Lemma 2.4. *[zigzag] Let $0 \to C_* \to D_* \to E_* \to 0$ be a short exact sequence of chain maps. For each p, there is a connecting homomorphism $\partial_* : H_p(E_*) \to H_{p-1}(C_*)$ such that the sequence*

$$\cdots \xrightarrow{\partial_*} C_p \to D_p \to E_p \xrightarrow{\partial_*} \cdots$$

is exact.

Proof. Read [Lee10], p.356. □

Theorem 2.5 (Naturality of the connecting homomorphism)**.** *supppose that the following diagram commutes:*

$$\begin{tikzcd}
0 \arrow[r] & C_* \arrow[r] \arrow[d, "\kappa"'] & D_* \arrow[r] \arrow[d, "\delta"] & E_* \arrow[r] \arrow[d, "\varepsilon"] & 0 \\
0 \arrow[r] & C'_* \arrow[r] & D'_* \arrow[r] & E'_* \arrow[r] & 0
\end{tikzcd}$$

with exact rows. Then the following diagram commutes for each p:

$$\begin{tikzcd}
H_p(E_*) \arrow[r, "\partial_*"'] \arrow[d, "\varepsilon_*"] & H_{p-1}(C_*) \arrow[d, "\kappa_*"'] \\
H_p(E'_*) \arrow[r, "\partial_*"] & H_{p-1}(C'_*)
\end{tikzcd}$$

Proof. Read [Lee10] p.359 □

3. Some Elementary Computations

Proposition 3.1. *Let $X = \bigsqcup_\alpha X_\alpha$ be decomposed according to its path components, and let $(i_\alpha)_* : X_\alpha \hookrightarrow X$ be the inclusion maps. Then, for each $p \geq 0$, the map*

$$\bigoplus_\alpha H_p(X_\alpha) \to H_p(X)$$

whose restriction to each $H_p(X_\alpha)$ is $(i_\alpha)*$ is an isomorphism.

Proof. exercise ☐

Corollary 3.2. *If X a topological space is decomposed as above, then $H_0(X)$ will be the free abelian group generated by the set of αs.*

Intuitively, any two points in a path component have a path connecting them, so every point in a path component is identified via a $1-$simplex i.e a path. So the generators of the group $H_0(X)$ will be represented by points, one from each path component. I.e, the group H_0 counts the number of path components.

Proposition 3.3. *If X is a discrete space, then $H_0(X)$ is the free abelian group on X, and for $p \geq 1$, we have that $H_p(X) = 0$*

Proof. The first claim follows directly from the previous corollary, and the fact that the only connected subsets of the discrete space are atmost singletons. Now, let $p \geq 1$. The we have for any singular $p-$simplex σ, that σ is a constant map, say with image $*$. Then

$$\partial\sigma = \sum_{i=0}^{p} (-1)^i \sigma \circ F_{i,p} = \begin{cases} 0 & \text{if } p \text{ is odd} \\ x \mapsto * & p \text{ even} \end{cases}$$

So if p is odd, then $\partial = 0$, and otherwise, it is an isomorphism. i.e The sequence is exact. Thus, for $p \geq 1$, $H_p(X) = 0$. ☐

4. Homotopy Invariance

As we have seen before, two spaces being homotopy equivalent implies that their fundamental groups are isomorphic. We see the same in the case of singular homology groups. Infact we have a slightly better result:

Theorem 4.1. *If $f_0, f_1 : X \to Y$ are homotopic maps, then for each $p \geq 0$, the induce homomorphisms $(f_0)_*, (f_1)_*$ are equal.*

Proof. I will not show the full proof here, one can read [Lee10] for that. But the proof is nice enough that I think providing atleast a sketch must be done. First we reduce the problem to the case $Y = X \times I$, $f_0(x) := i_0(x) = x \mapsto (x, 0)$ and $f_1(x) = i_1(x) := x \mapsto (x, 1)$, by simply using the functoriality of taking the fundamental group. Then we note that in order to show this, it suffices to find homomorphisms $h : C_p(X) \to C_{p+1}(X \times I)$ such that

$$h \circ \partial + \partial \circ h = (i_1)_\sharp - (i_0)_\sharp.$$

This is simply because for any boundary ∂c, we have $h \circ \partial(\partial c) + \partial \circ h(\partial c) = \partial \circ h(\partial c) \in \ker(\partial)$.

the key fact used to construct h is that a $p-$prism can be constructed as a sum of some $p-$simplies. So the term $\partial \circ h$ constructs the bonudary of the prism, and the term $h \circ \partial$ cancels out the sides, giving us the difference of the top and the bottom. Lee has a nice graphic of this in his book [Lee10]. ☐

Corollary 4.2. *Let X be a contractible space. Then $H_0(X) = \mathbb{Z}, H_p(X) = 0 \quad \forall p \geq 1$.*

5. Mayer-Vietoris Theorem

It turns out that for calculating homology groups, a result analogous to the Seifert-Van Kampen theorem exists. We begin by investigating the same diagram that we investigate for the Van Kampen theorem, but from the different viewpoint.

Let $X = U \cup V$ be a union of two open subsets. We have inclusion maps

$$\begin{tikzcd} & U \arrow[dr, "k"] & \\ U \cap V \arrow[ur, "i"] \arrow[dr, "j"'] & & X \\ & V \arrow[ur, "l"'] & \end{tikzcd}$$

and we can "take the homology" of this diagram, to get the corresponding diagram of homology groups. The Mayer-Vietoris theorem says the following:

Theorem 5.1 (Mayer-Vietoris)**.** *Let X be a topological space, and let U, V, be open subsets whose union is X. Then for each p there is a homomorphism $\partial_* : H_p(X) \to H_{p-1}(U \cap V)$ such that the following sequence is exact:*

$$\cdots \xrightarrow{\partial_*} H_p(U \cap V) \xrightarrow{i_* \oplus j_*} H_p(U) \oplus H_p(V) \xrightarrow{k_* - l_*} H_p(X)$$
$$\xrightarrow{\partial_*} H_{p-1}(U \cap V) \xrightarrow{i_* \oplus j_*} \cdots$$

Proof. Read [Lee10] ☐

6. Homology Groups of Spheres

We use the results in the previous section to compute homology group of spheres.

Theorem 6.1 (Homology Groups of Sphere)**.** *For $n \geq 1$, $\mathbb{S}^n$ has the following singular homology groups:*

$$H_p(\mathbb{S}^n) = \begin{cases} \mathbb{Z} & \text{if } p = 0 \\ 0 & \text{if } 0 < p < n \\ \mathbb{Z} & \text{if } p = n \\ 0 & \text{if } p > n \end{cases}$$

Proof. We use the Mayer-Vietoris sequence as follows. Let N, S denote the north and south poles , and let $U = \mathbb{S}^n \backslash \{N\}, V = \mathbb{S}^n \backslash \{S\}$. Mayer-Vietoris gives us:

$$H_p(U) \oplus H_p(V) \to H_p(\mathbb{S}^n) \xrightarrow{\partial_*} H_{p-1}(U \cap V) \to H_{p-1}(U) \oplus H_{p-1}(V).$$

Since U, V are contractible, when $p > 1$, we have that the following sequence is exact:

$$0 \to H_p(\mathbb{S}^n) \xrightarrow{\partial_*} H_{p-1}(U \cap V) \to 0.$$

But $U \cap V$ is homotopy equivalent to $\mathbb{S}^{n-1}$, so we have $H_p(\mathbb{S}^n) \cong H_{p-1}(\mathbb{S}^{n-1})$. By induction, we are done. ☐

7. Degree Theory

Suppose that $n \geq 1$, and let $f : \mathbb{S}^n \to \mathbb{S}^n$ be continuous. The map $f_* : H_n(\mathbb{S}^n) \to H_n(\mathbb{S}^n)$, is a homomorphism $\mathbb{Z} \to \mathbb{Z}$, which corresponds to multiplication by an integer, called the degree of f.

Proposition 7.1. *Suppose $n \geq 1$, and $f, g : \mathbb{S}^n \to \mathbb{S}^n$ are continuous maps.*

1. $\deg(f \circ g) = \deg f \deg g$.
2. $f \simeq g$, *then* $\deg f = \deg g$.

Proof. Part 1 follows from the fact that taking homologies is a functor, and part 2 from the homotopy invariance of singular homology. □

Theorem 7.2. *Let $\varphi : \mathbb{S}^n \to \mathbb{S}^n$ be continupus. $\deg \varphi \neq 0$ implies that φ is surjective. If $\deg \varphi \neq (-1)^{n+1}$, then φ has a fixed point.*

Proof. Indeed, for the first statement, if the image of φ was not all of $\mathbb{S}^n$, then the map φ would be homotopic to a constant map. This is because if the image misses a point p, then $\mathbb{S}^n \backslash \{p\}$ is contractible. Thus the map φ induces a trivial homomorphism which has degreee 0.
Now suppose that f does not have any fixed points. Let $a : \mathbb{S}^n \to \mathbb{S}^n$ denote the antipodal map. Then $a \circ f \simeq id$. This is because the image of any point will not be $-x$, so the image $a(f(x))$ can be traced back to x via a homotopy on a great circle. This, $\deg a \deg f = \deg id \implies \deg f = (-1)^{n+1}$ □

Theorem 7.3. *The antipodal map $\alpha : \mathbb{S}^n \to \mathbb{S}^n$ is homotopic to the identity iff n is odd.*

Proof. If $n = 2k - 1$ is odd, then an explicit homotopy is given by

$$H(x,t) = (\cos \pi t \cdot x_1 + \sin \pi t \cdot x_2, \cos \pi t \cdot x_2 - \sin \pi t \cdot x_1, \cdots, \cos \pi t \cdot x_{2k} - \sin \pi t \cdot x_{2k-1})$$

If $n = 0$, then *alpha* is clearly not homotopic to the identity. If $n > 0$ is even, then $\deg \alpha = -1$, but $\deg id = 1$ thus proving the theorem. □

Theorem 7.4 (Hairy Ball)**.** *There exists a nowhere vanishing vector field on $\mathbb{S}^n$ iff n is odd.*

Proof. Suppose that there exists such a vector field. Then we can use $V/|V|$ to construct a homotopy between the antipodal map α and id as follows.

$$H(x,t) = (\cos \pi t)x + (\sin \pi t)V(x)/|V(x)|.$$

Since $V(x) \cdot x = 0$ by definition of a vector field and $|x|^2 = |V(x)|^2 = 1$, we get that the above map is the desired homotopy. Therefore n must be odd. □

Bibliography

[Lee10] John M Lee. *Introduction to Topological Manifolds.* Graduate Texts in Mathematics. Springer, New York, NY, 2 edition, December 2010. 7, 11, 12, 13